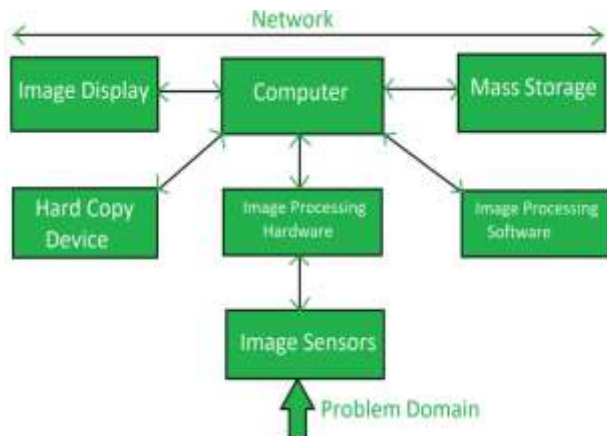
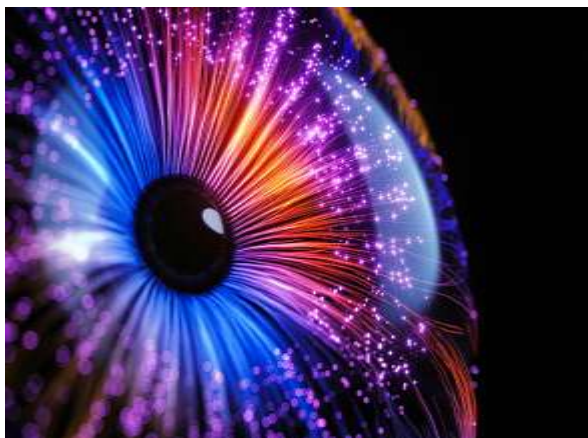


Digital Image Processing



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Topic 9

Image Segmentation



Introduction

Topics:

Line Detection

Edge Detection

Local Processing

Regional Processing

Region based Segmentation

Watershed Segmentation

Clustering



Introduction

Image segmentation is a computer vision technique that partitions an image into multiple segments, or regions, to identify objects and boundaries, enabling detailed image analysis and processing. It works by classifying individual pixels and grouping them into meaningful segments based on features like color or brightness, often using **deep learning models** trained on annotated datasets. Key applications include medical imaging for disease diagnosis, self-driving car technology, and robotics.



Introduction

How it Works

- 1. Pixel Analysis:** The process involves a computer vision model that examines each pixel in an image.
- 2. Feature Extraction:** The model uses visual features such as color, brightness, and texture to classify pixels and identify object boundaries.
- 3. Segmentation:** Pixels are grouped into discrete segments, creating a partitioned image that highlights specific regions or objects.
- 4. Output:** The output is an array of classified pixels, representing the segmented areas within the image.



Applications

- **Medical Imaging:** Used to detect and segment organs, tumors, or other anomalies in medical scans for faster and more accurate diagnoses.
- **Self-Driving Cars:** Helps in identifying and tracking objects like roads, vehicles, and pedestrians, crucial for autonomous navigation.
- **Robotics:** Enables robots to understand their environment by segmenting objects for interaction and manipulation.
- **Satellite Imagery:** Used to identify and map features in satellite images, such as land use or specific objects.
- **Image Editing:** Allows for the separation of objects from their backgrounds, enabling tasks like image background removal.



Types of Image Segmentation

Image segmentation techniques can be broadly categorized into semantic segmentation, instance segmentation and panoptic segmentation. These methods aim to divide an image into meaningful regions, but differ in their approach to object identification and distinction.

1. **Semantic Segmentation:** It involves assigning a class label to every pixel in an image based on shared characteristics such as color, texture and shape. This method treats all pixels belonging to the same class as identical, without distinguishing between individual objects. For example in an image with multiple trees all pixels corresponding to any tree would be labeled as "tree" regardless of how many trees appear in the image.
2. **Instance segmentation:** It extends semantic segmentation by not only labeling color of each pixel but also distinguishing between individual objects of the same class. This approach identifies each object of the same class as a unique instance. For example if there are multiple cats in an image it will differentiate each cat and treat them as separate entities despite them all being labeled as "cats."



3. **Panoptic Segmentation:** It combines both semantic and instance segmentation techniques providing a complete image analysis. It assigns a class label to every pixel and also detects individual objects. This combined approach helps us to understand both broad categories and detailed object boundaries simultaneously. For example in a traffic scene it would label all pedestrians and cars (semantic segmentation) while also outlining the location of each individual person and car (instance segmentation).



Techniques

Thresholding: Thresholding method involves selecting a threshold value and classifying image pixels between foreground and background based on intensity values.

Edge Detection: It identifies abrupt change in intensity or discontinuation in the image. It uses algorithms like Sobel, Canny or Laplacian edge detectors.

Region based segmentation: This method segments the image into smaller regions and iteratively merges them based on predefined attributes in color, intensity and texture to handle noise and irregularities in the image.

Clustering Algorithm: This method uses algorithms like K-means or Gaussian models to group object pixels in an image into clusters based on similar features like color or texture.

Watershed Segmentation: It treats the image like a topographical map where the watershed lines are identified based on pixel intensity and connectivity like water flowing down different valleys.



Deep Learning Based Models

U-Net: Designed for medical image segmentation, it uses a U-shaped architecture that allows precise segmentation even with limited data. It efficiently captures context while retaining fine details through skip connections making it ideal for small, high-resolution images.

Fully Convolutional Network (FCN): It modify conventional CNNs by replacing fully connected layers with convolutional layers allowing the network to process images of any size. This design enables pixel-wise segmentation, outputting spatial maps of segmentation results.

SegNet: With its encoder-decoder structure, it is highly effective for scene understanding and object recognition. The encoder captures contextual information from the image while the decoder uses this context to perform precise localization and segmentation.

Mask R-CNN: An extension of Faster R-CNN for object detection, it adds a branch to predict segmentation masks alongside bounding box coordinates. It is used for instance segmentation, identifying and delineating individual object instances in images.

Vision Transformer (ViT): ViT applies transformer architectures to image segmentation by dividing an image into patches and processing them sequentially. This allows the model to capture global context, enhancing segmentation accuracy for complex visual tasks.

ResNet (Residual Network): While ResNet itself is a classification model, its residual learning framework is widely used as a backbone in many segmentation models like DeepLab, Mask R-CNN. Residual blocks help in training very deep networks by solving the vanishing gradient problem, enabling better feature extraction and improved segmentation accuracy.



Challenges

Complex backgrounds: Segmentation can struggle when the background is complex or there's little contrast between objects and the surroundings.

Variety in object appearance: Objects might look different in various lighting conditions making it harder to segment consistently.

Computational cost: Advanced methods like deep learning can require significant computational power and time.

Overlapping Objects: When objects overlap or are partially occluded distinguishing between them can be difficult, leading to poor segmentation accuracy.



a	b	c
d	e	f

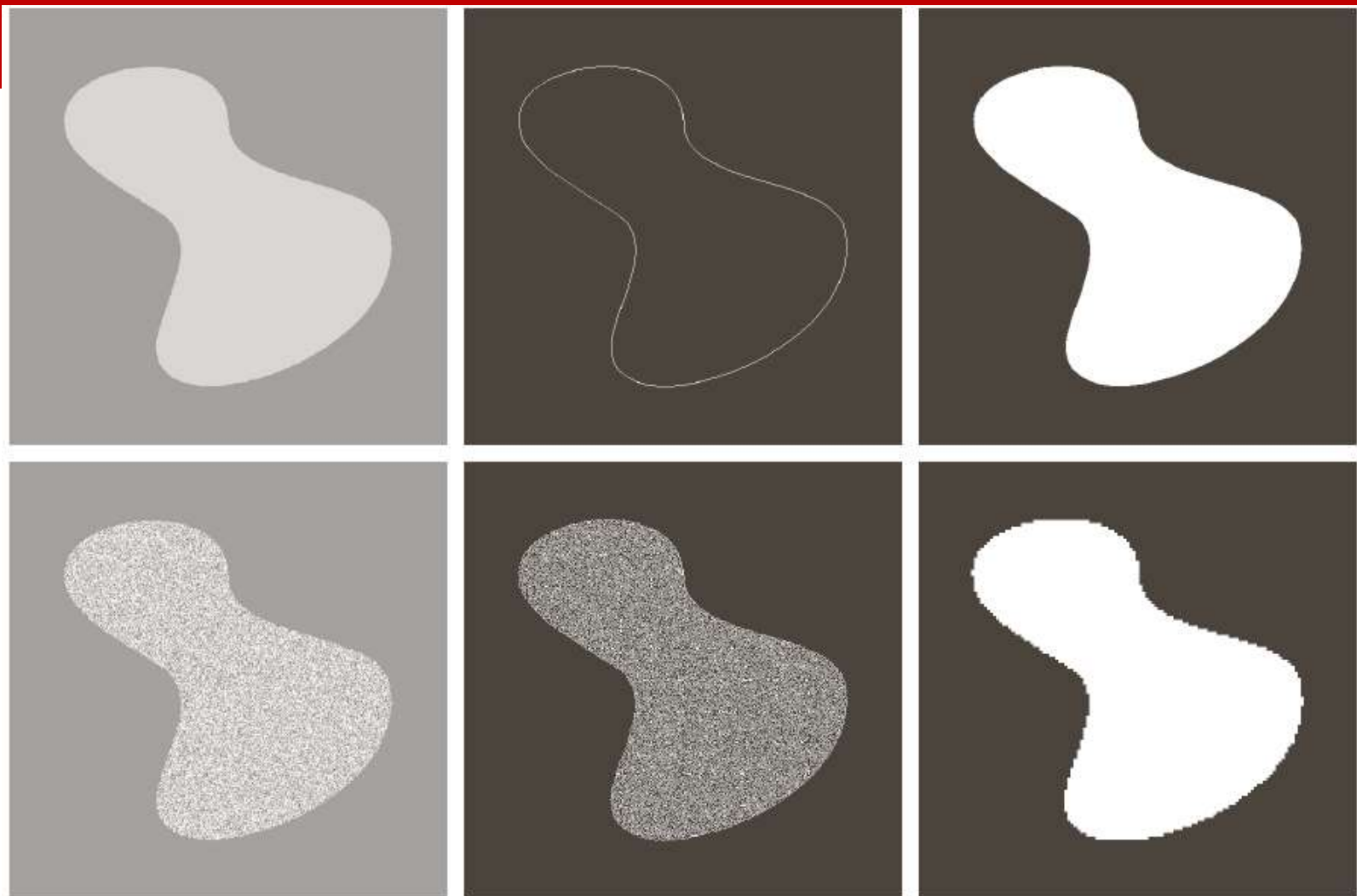


FIGURE 10.1 (a) Image containing a region of constant intensity. (b) Image showing the boundary of the inner region, obtained from intensity discontinuities. (c) Result of segmenting the image into two regions. (d) Image containing a textured region. (e) Result of edge computations. Note the large number of small edges that are connected to the original boundary, making it difficult to find a unique boundary using only edge information. (f) Result of segmentation based on region properties.



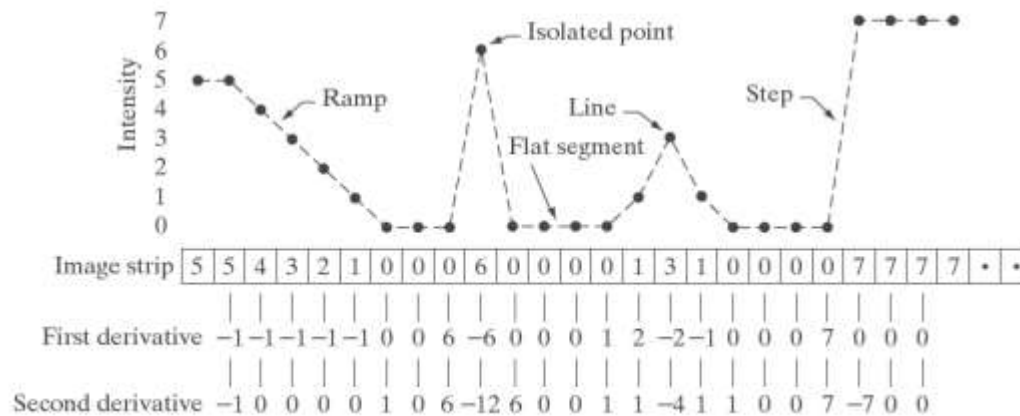
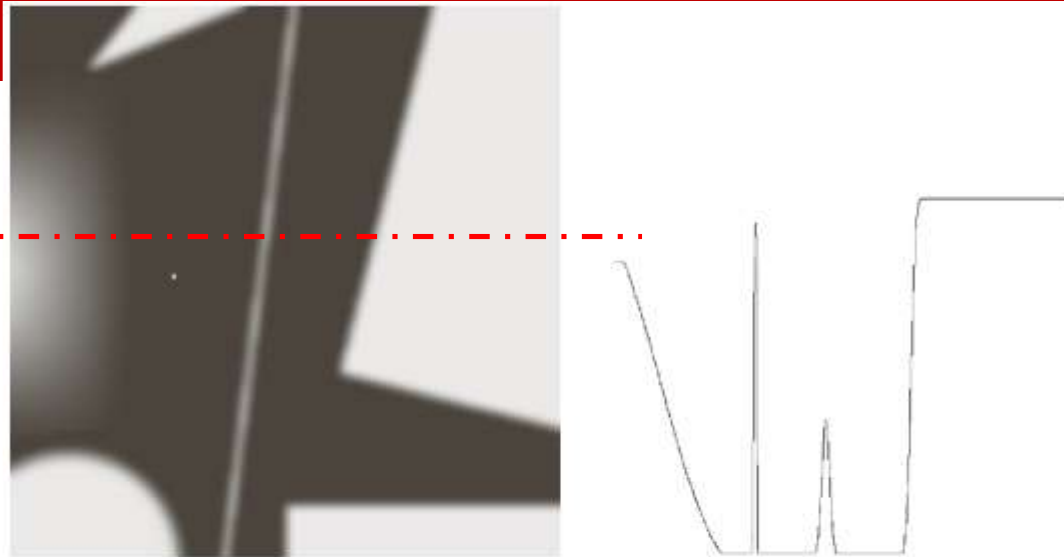
Background

- First-order derivative

$$\frac{\partial f}{\partial x} = f'(x) = f(x+1) - f(x)$$

- Second-order derivative

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



a b
c

FIGURE 10.2 (a) Image. (b) Horizontal intensity profile through the center of the image, including the isolated noise point. (c) Simplified profile (the points are joined by dashes for clarity). The image strip corresponds to the intensity profile, and the numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10.2-1) and (10.2-2).



Characteristics of First and Second Order Derivatives

- First-order derivatives generally produce thicker edges in image
- Second-order derivatives have a stronger response to fine detail, such as thin lines, isolated points, and noise
- Second-order derivatives produce a double-edge response at ramp and step transition in intensity
- The sign of the second derivative can be used to determine whether a transition into an edge is from light to dark or dark to light



Detection of Isolated Points

- The Laplacian

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

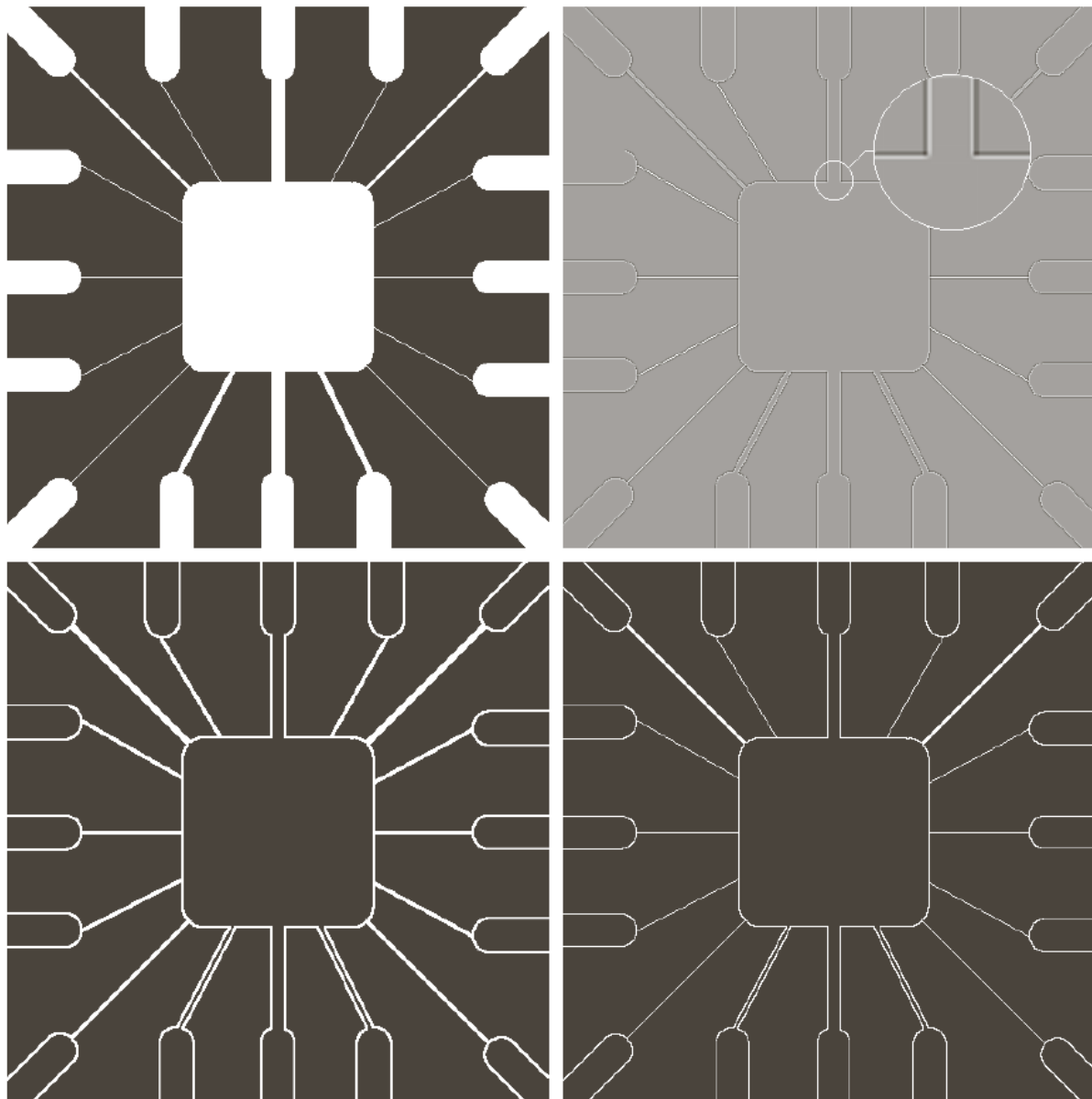
$$= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

$$g(x, y) = \begin{cases} 1 & \text{if } |R(x, y)| \geq T \\ 0 & \text{otherwise} \end{cases} \quad R = \sum_{k=1}^9 w_k z_k$$



Line Detection

- Second derivatives to result in a stronger response and to produce thinner lines than first derivatives
- Double-line effect of the second derivative must be handled properly



a	b
c	d

FIGURE 10.5

(a) Original image.

(b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian.

(c) Absolute value of the Laplacian.

(d) Positive values of the Laplacian.

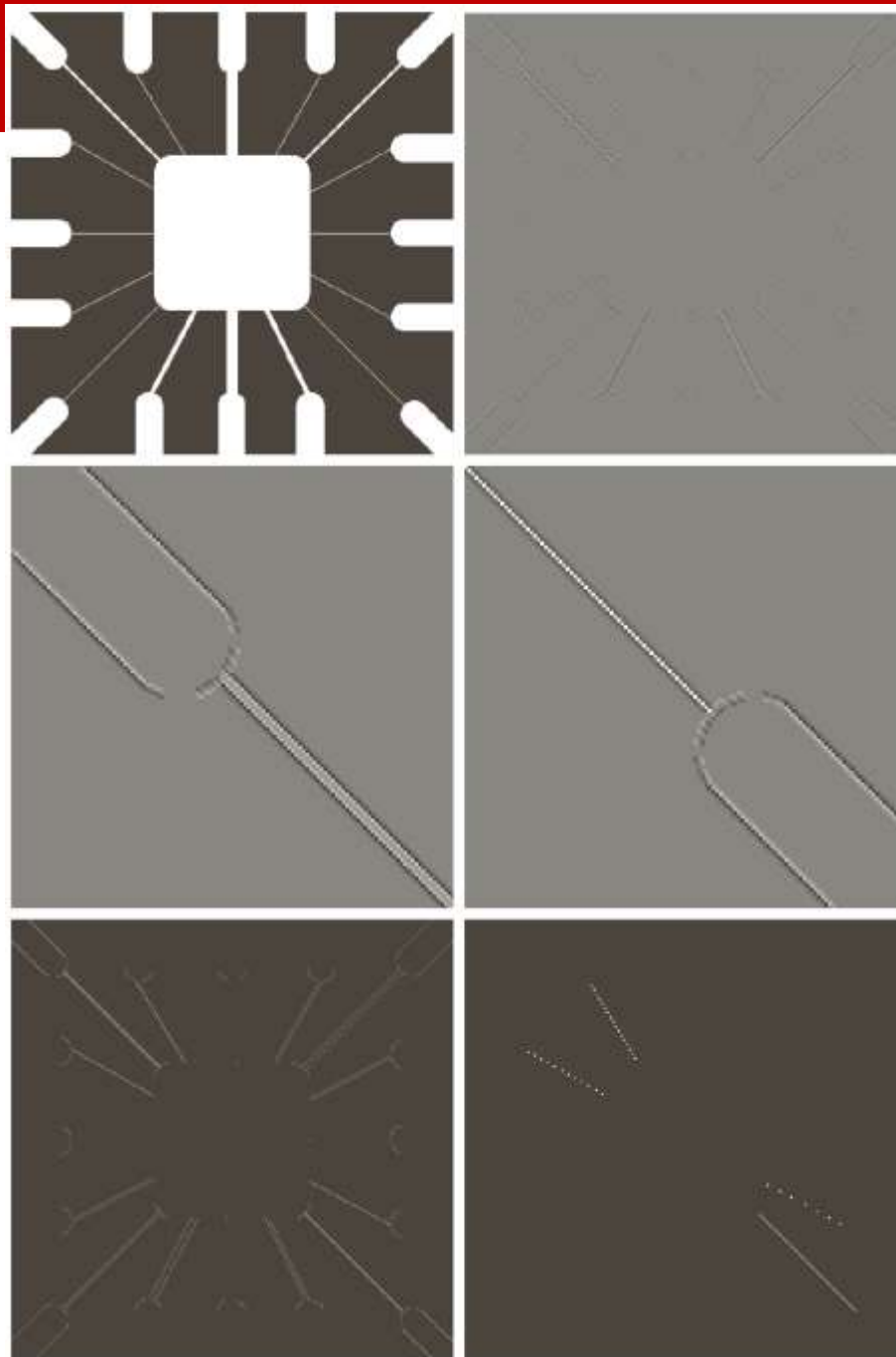


Detecting Line in Specified Directions

-1	-1	-1	2	-1	-1	-1	2	-1	-1	-1	2
2	2	2	-1	2	-1	-1	2	-1	-1	2	-1
-1	-1	-1	-1	-1	2	-1	2	-1	2	-1	-1
Horizontal			+45°			Vertical			-45°		

FIGURE 10.6 Line detection masks. Angles are with respect to the axis system in Fig. 2.18(b).

- Let R_1 , R_2 , R_3 , and R_4 denote the responses of the masks in Fig. 10.6. If, at a given point in the image, $|R_k| > |R_j|$, for all $j \neq k$, that point is said to be more likely associated with a line in the direction of mask k .



a	b
c	d
e	f

FIGURE 10.7

(a) Image of a wire-bond template.

(b) Result of processing with the $+45^\circ$ line detector mask in Fig. 10.6.

(c) Zoomed view of the top left region of (b).

(d) Zoomed view of the bottom right region of (b).

(e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g \geq T$, where g is the image in (e). (The points in (f) were enlarged to make them easier to see.)



Edge Detection

- Edges are pixels where the brightness function changes abruptly
- Edge models

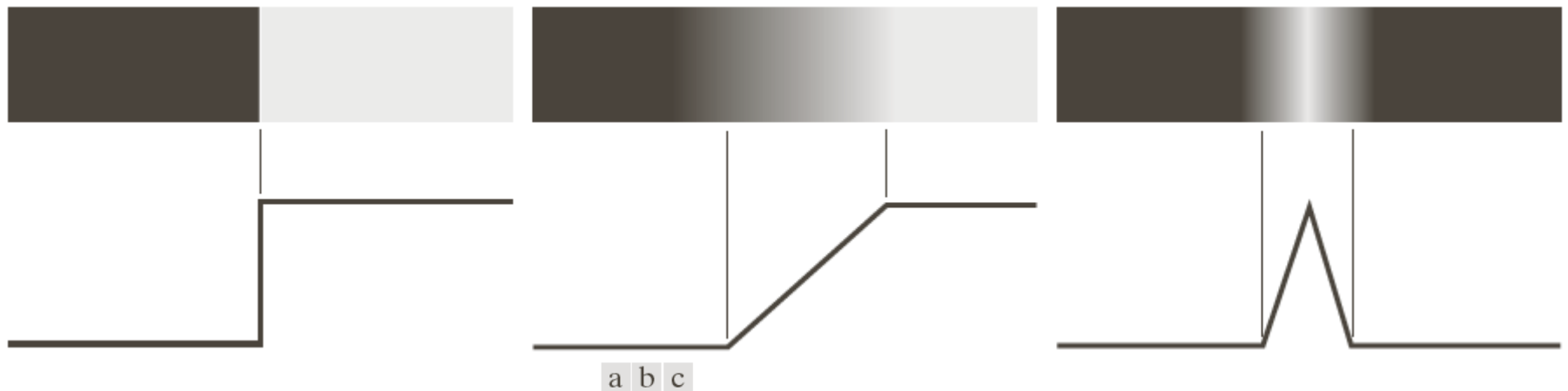
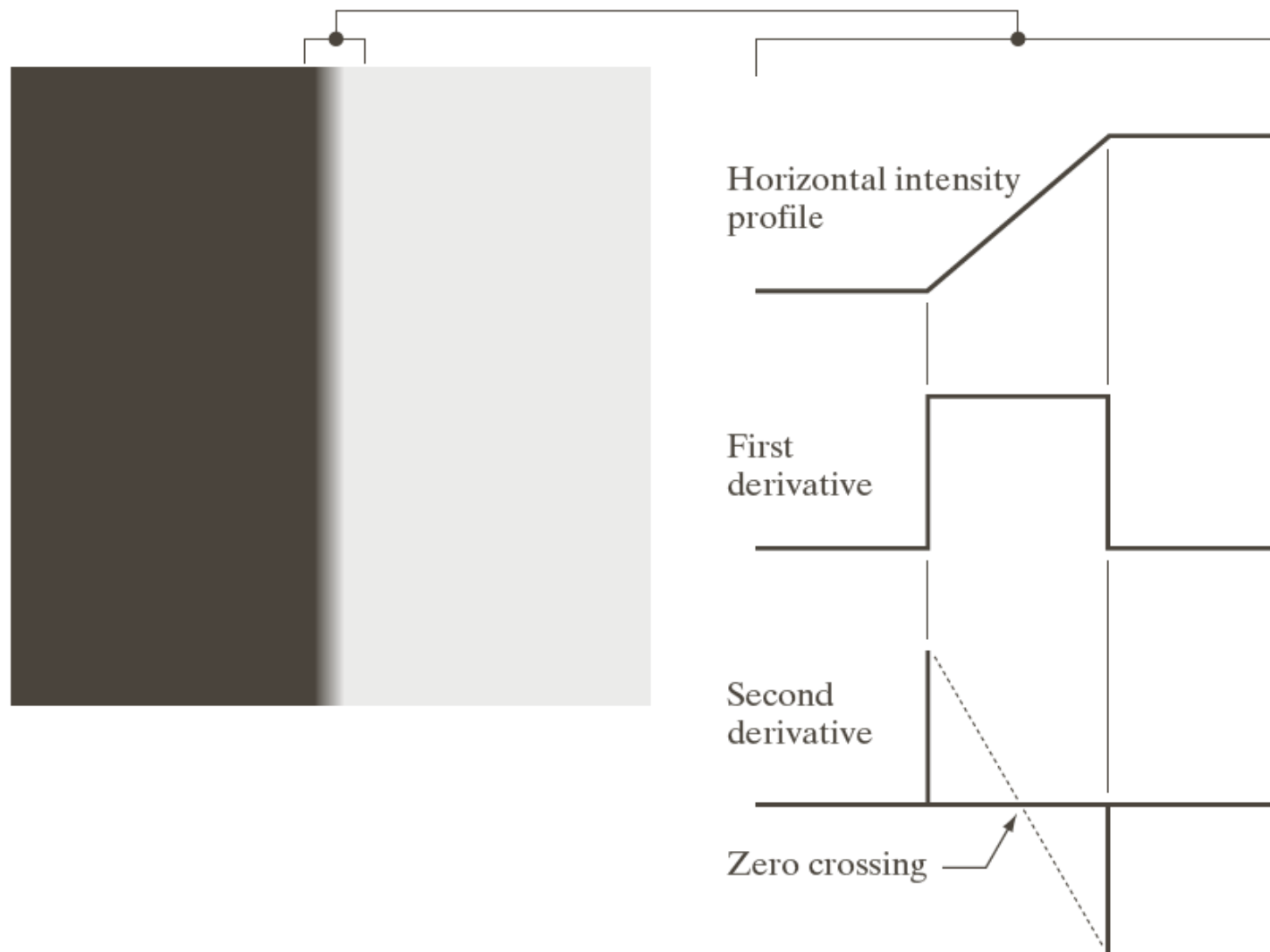


FIGURE 10.8

From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.



a b

FIGURE 10.10
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.



Basic Edge Detection by Using First- Order Derivative

$$\nabla f \equiv \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

The magnitude of ∇f

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

The direction of ∇f

$$\alpha(x, y) = \tan^{-1} \left[\frac{g_x}{g_y} \right]$$

The direction of the edge

$$\phi = \alpha - 90^\circ$$



$$\text{Edge normal: } \nabla f \equiv \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\text{Edge unit normal: } \nabla f / \text{mag}(\nabla f)$$

In practice, sometimes the magnitude is approximated by

$$\text{mag}(\nabla f) = \left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right| \text{ or } \text{mag}(\nabla f) = \max \left(\left| \frac{\partial f}{\partial x} \right|, \left| \frac{\partial f}{\partial y} \right| \right)$$

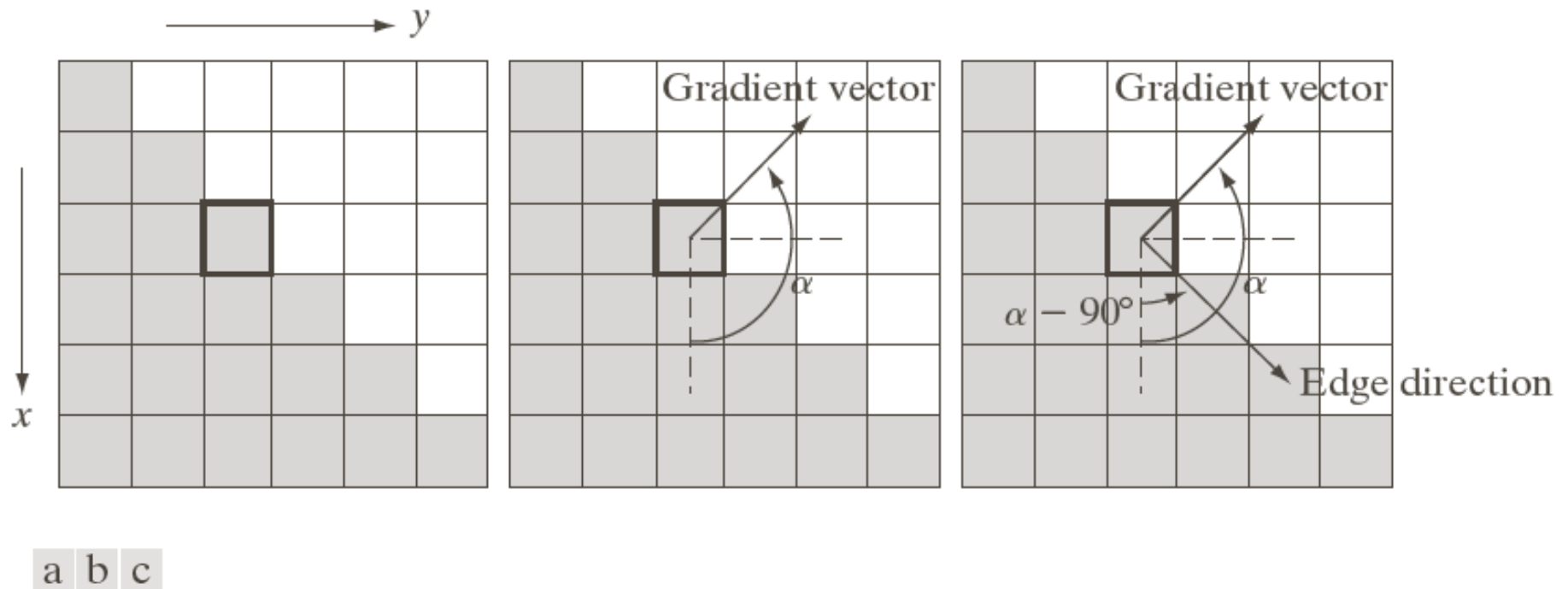


FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.



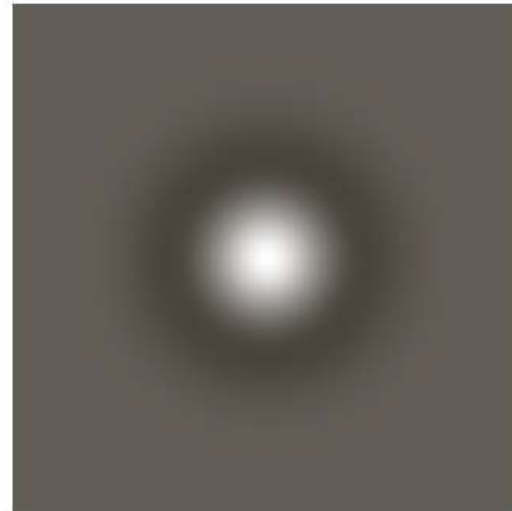
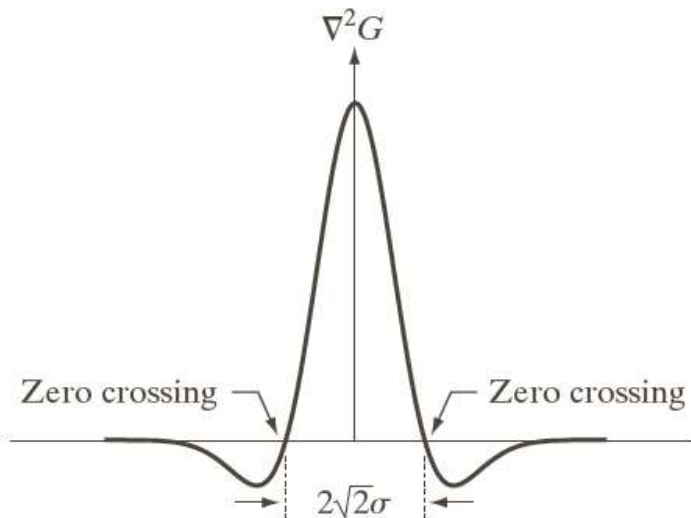
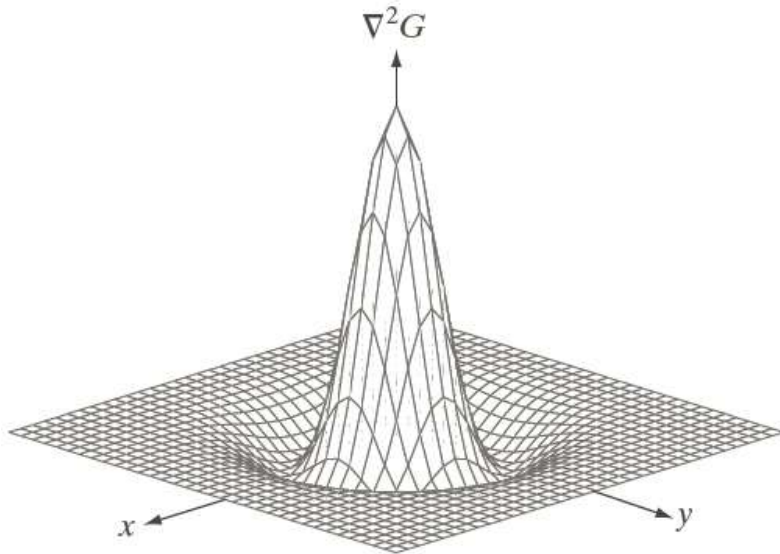
Advanced Techniques for Edge Detection

The Marr-Hildreth edge detector

$$G(x, y) = e^{-\frac{x^2+y^2}{2\sigma^2}}, \quad \sigma : \text{space constant.}$$

Laplacian of Gaussian (LoG)

$$\begin{aligned}\nabla^2 G(x, y) &= \frac{\partial^2 G(x, y)}{\partial x^2} + \frac{\partial^2 G(x, y)}{\partial y^2} \\&= \frac{\partial}{\partial x} \left[\frac{-x}{\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \right] + \frac{\partial}{\partial y} \left[\frac{-y}{\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \right] \\&= \left[\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2} \right] e^{-\frac{x^2+y^2}{2\sigma^2}} + \left[\frac{y^2}{\sigma^4} - \frac{1}{\sigma^2} \right] e^{-\frac{x^2+y^2}{2\sigma^2}} \\&= \left[\frac{x^2 + y^2 - \sigma^2}{\sigma^4} \right] e^{-\frac{x^2+y^2}{2\sigma^2}}\end{aligned}$$



0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

a b
c d

FIGURE 10.21

(a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5×5 mask approximation to the shape in (a). The negative of this mask would be used in practice.



Marr-Hildreth Algorithm

1. Filter the input image with an $n \times n$ Gaussian lowpass filter.
 n is the smallest odd integer greater than or equal to 6
2. Compute the Laplacian of the image resulting from step 1
3. Find the zero crossing of the image from step 2

$$g(x, y) = \nabla^2 [G(x, y) \star f(x, y)]$$



The Canny Edge Detector

- **Optimal for step edges corrupted by white noise.**
- **The Objective**
 1. **Low error rate**

The edges detected must be as close as possible to the true edge
 2. **Edge points should be well localized**

The edges located must be as close as possible to the true edges
 3. **Single edge point response**

The number of local maxima around the true edge should be minimum



Let $f(x, y)$ denote the input image and $G(x, y)$ denote the Gaussian function:

$$G(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

We form a smoothed image, $f_s(x, y)$ by convolving G and f :

$$f_s(x, y) = G(x, y) \star f(x, y)$$



Compute the gradient magnitude and direction (angle):

$$M(x, y) = \sqrt{g_x^2 + g_y^2}$$

and

$$\alpha(x, y) = \arctan(g_y / g_x)$$

where $g_x = \partial f_s / \partial x$ and $g_y = \partial f_s / \partial y$

Note: any of the filter mask pairs in Fig.10.14 can be used to obtain g_x and g_y

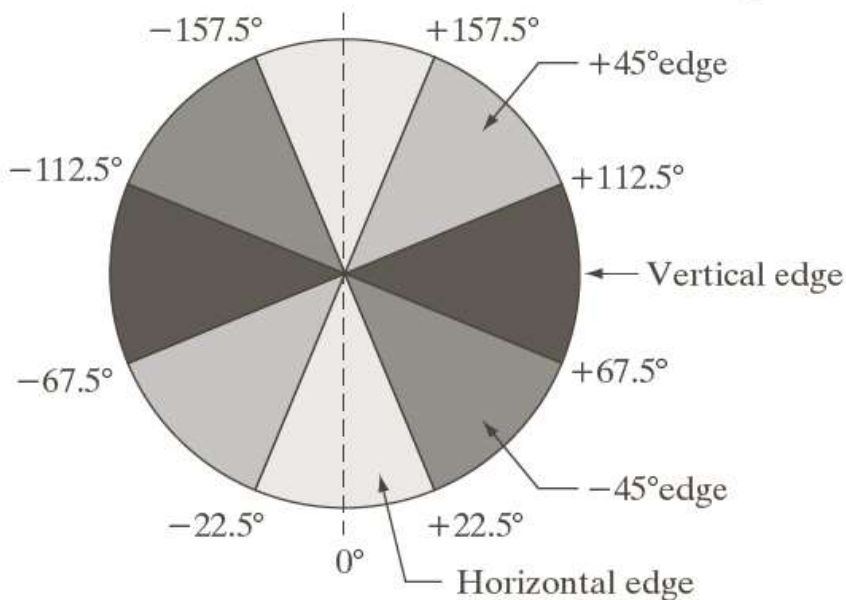
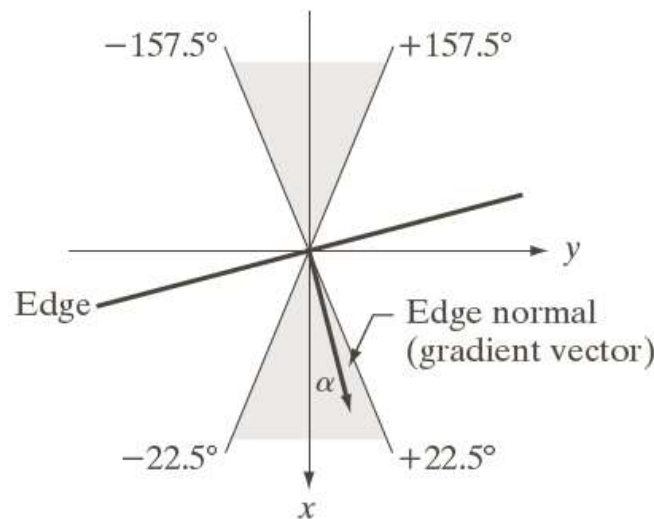
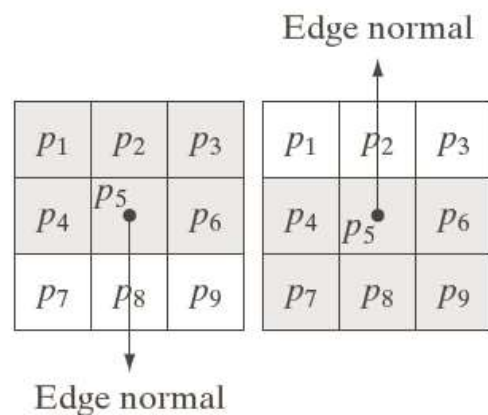


The gradient $M(x, y)$ typically contains wide ridge around local maxima. Next step is to thin those ridges.

Nonmaxima suppression:

Let d_1, d_2, d_3 , and d_4 denote the four basic edge directions for a 3×3 region: horizontal, -45° , vertical, $+45^\circ$, respectively.

1. Find the direction d_k that is closest to $\alpha(x, y)$.
2. If the value of $M(x, y)$ is less than at least one of its two neighbors along d_k , let $g_N(x, y) = 0$ (suppression); otherwise, let $g_N(x, y) = M(x, y)$



a b
c

FIGURE 10.24

(a) Two possible orientations of a horizontal edge (in gray) in a 3×3 neighborhood.

(b) Range of values (in gray) of α , the direction angle of the *edge normal*,

for a horizontal edge. (c) The angle ranges of the edge normals for the four types of edge directions in a 3×3 neighborhood.

Each edge direction has two ranges, shown in corresponding shades of gray.



The final operation is to threshold $g_N(x, y)$ to reduce false edge points.

Hysteresis thresholding:

$$g_{NH}(x, y) = g_N(x, y) \geq T_H$$

$$g_{NL}(x, y) = g_N(x, y) \geq T_L$$

and

$$g_{NL}(x, y) = g_{NL}(x, y) - g_{NH}(x, y)$$



Depending on the value of T_H , the edges in $g_{NH}(x, y)$ typically have gaps. Longer edges are formed using the following procedure:

- (a). Locate the next unvisited edge pixel, p , in $g_{NH}(x, y)$.
- (b). Mark as valid edge pixel all the weak pixels in $g_{NL}(x, y)$ that are connected to p using 8-connectivity.
- (c). If all nonzero pixel in $g_{NH}(x, y)$ have been visited go to step (d), else return to (a).
- (d). Set to zero all pixels in $g_{NL}(x, y)$ that were not marked as valid edge pixels.



The Canny Edge Detection: Summary

- Smooth the input image with a Gaussian filter
- Compute the gradient magnitude and angle images
- Apply nonmaxima suppression to the gradient magnitude image
- Use double thresholding and connectivity analysis to detect and link edges



Edge Linking and Boundary Detection

- Edge detection typically is followed by linking algorithms designed to assemble edge pixels into meaningful edges and/or region boundaries
- Three approaches to edge linking
 - Local processing
 - Regional processing
 - Global processing



Local Processing

- Analyze the characteristics of pixels in a small neighborhood about every point (x,y) that has been declared an edge point
- All points that similar according to predefined criteria are linked, forming an edge of pixels.

Establishing similarity: (1) the strength (magnitude) and (2) the direction of the gradient vector.

A pixel with coordinates (s,t) in S_{xy} is linked to the pixel at (x,y) if both magnitude and direction criteria are satisfied.



Let S_{xy} denote the set of coordinates of a neighborhood centered at point (x, y) in an image. An edge pixel with coordinate (s, t) in S_{xy} is similar in *magnitude* to the pixel at (x, y) if

$$|M(s, t) - M(x, y)| \leq E$$

An edge pixel with coordinate (s, t) in S_{xy} is similar in *angle* to the pixel at (x, y) if

$$|\alpha(s, t) - \alpha(x, y)| \leq A$$



1. Compute the gradient magnitude and angle arrays, $M(x,y)$ and $\alpha(x,y)$ of the input image $f(x,y)$
2. Form a binary image, g , whose value at any pair of coordinates (x,y) is given by

$$g(x,y) = \begin{cases} 1 & \text{if } M(x,y) > T_M \text{ and } \alpha(x,y) = A \pm T_A \\ 0 & \text{otherwise} \end{cases}$$

T_M : threshold A : specified angle direction

T_A : a "band" of acceptable directions about A



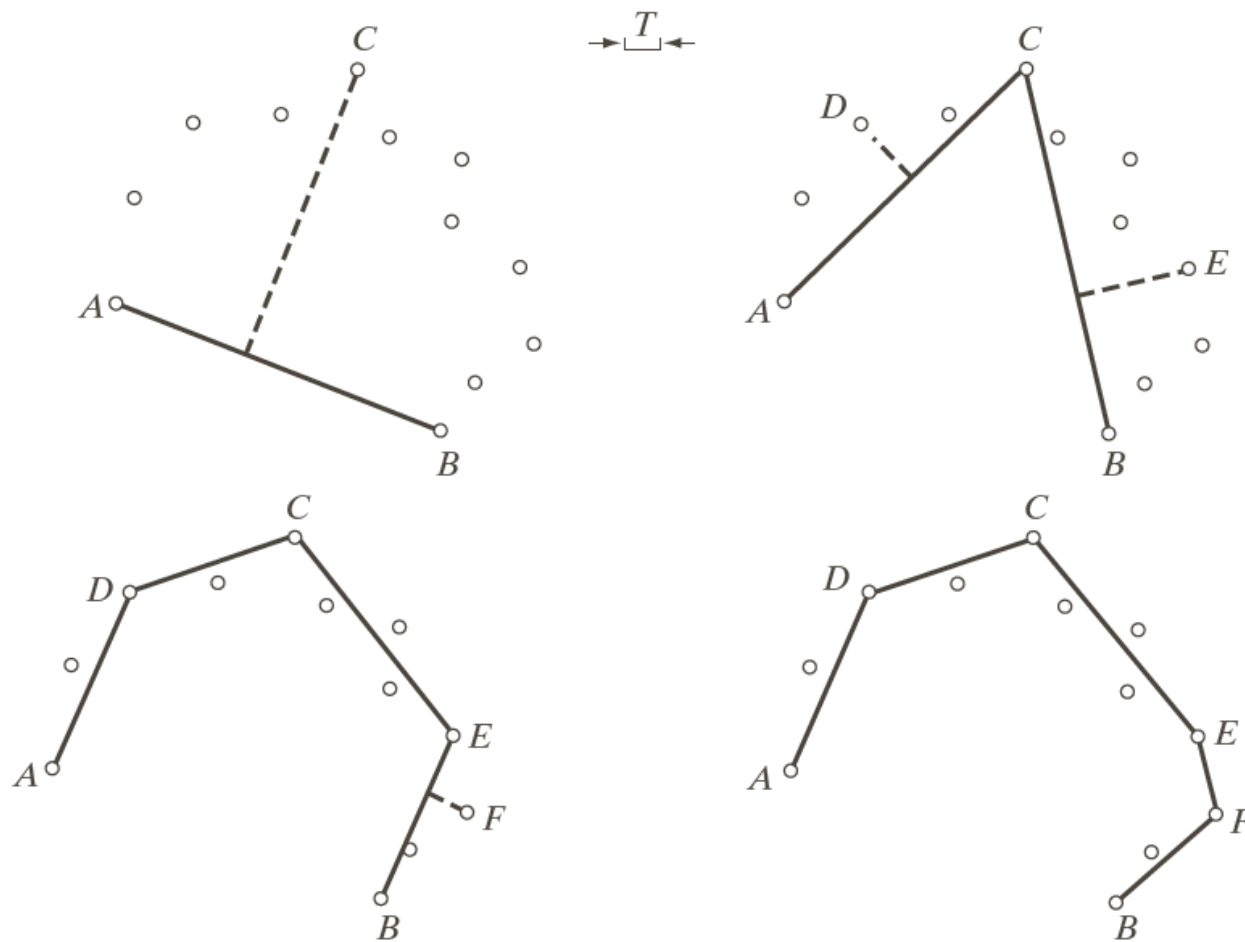
3. Scan the rows of g and fill (set to 1) all gaps (sets of 0s) in each row that do not exceed a specified length, K .
4. To detect gaps in any other direction, rotate g by this angle and apply the horizontal scanning procedure in step 3.



Regional Processing

- The location of regions of interest in an image are known or can be determined
- Polygonal approximations can capture the essential shape features of a region while keeping the representation of the boundary relatively simple
- Open or closed curve

Open curve: a large distance between two consecutive points in the ordered sequence relative to the distance between other points



a	b
c	d

FIGURE 10.28
Illustration of the
iterative
polygonal fit
algorithm.



1. Let P be the sequence of ordered, distinct, 1-valued points of a binary image. Specify two starting points, A and B .
2. Specify a threshold, T , and two empty stacks, $OPEN$ and $CLOSED$.
3. If the points in P correspond to a closed curve, put A into $OPEN$ and put B into $OPEN$ and $CLOSES$. If the points correspond to an open curve, put A into $OPEN$ and B into $CLOSED$.
4. Compute the parameters of the line passing from the last vertex in $CLOSED$ to the last vertex in $OPEN$.



5. Compute the distances from the line in Step 4 to all the points in P whose sequence places them between the vertices from Step 4. Select the point, $V_{\max}$, with the maximum distance, $D_{\max}$
6. If $D_{\max} > T$, place $V_{\max}$ at the end of the OPEN stack as a new vertex. Go to step 4.
7. Else, remove the last vertex from OPEN and insert it as the last vertex of CLOSED.
8. If OPEN is not empty, go to step 4.
9. Else, exit. The vertices in CLOSED are the vertices of the polygonal fit to the points in P .



Region-Based Segmentation

- **Region Growing**

1. Region growing is a procedure that groups pixels or subregions into larger regions.
2. The simplest of these approaches is *pixel aggregation*, which starts with a set of “seed” points and from these grows regions by appending to each seed points those **neighboring pixels** that have **similar properties** (such as gray level, texture, color, shape).
3. Region growing based techniques are better than the edge-based techniques in noisy images where edges are difficult to detect.



Example: **Region Growing based on 8-connectivity**

$f(x, y)$: input image array

$S(x, y)$: seed array containing 1s (seeds) and 0s

$Q(x, y)$: predicate



1. Find all connected components in $S(x, y)$ and erode each connected components to one pixel; label all such pixels found as 1. All other pixels in S are labeled 0.
2. Form an image f_Q such that, at a pair of coordinates (x, y) , let $f_Q(x, y) = 1$ if the Q is satisfied otherwise $f_Q(x, y) = 0$.
3. Let g be an image formed by appending to each seed point in S all the 1-value points in f_Q that are 8-connected to that seed point.
4. Label each connencted component in g with a different region label. This is the segmented image obtained by region growing.



$$Q = \begin{cases} \text{TRUE} & \text{if the absolute difference of the intensities} \\ & \text{between the seed and the pixel at (x,y) is } \leq T \\ \text{FALSE} & \text{otherwise} \end{cases}$$



Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10



Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

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59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

4-connectivity



Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

10	10	10	10	10	10	10
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10	59	10	60	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

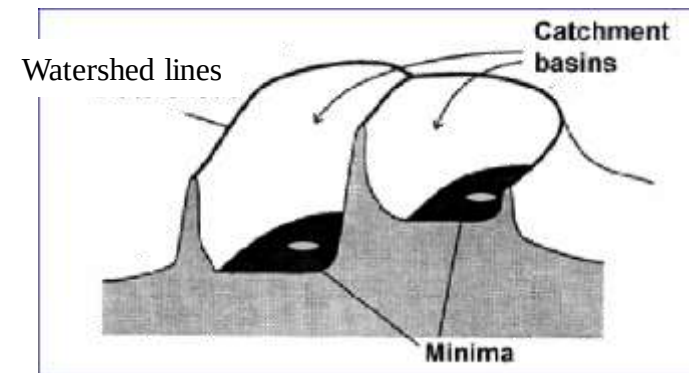
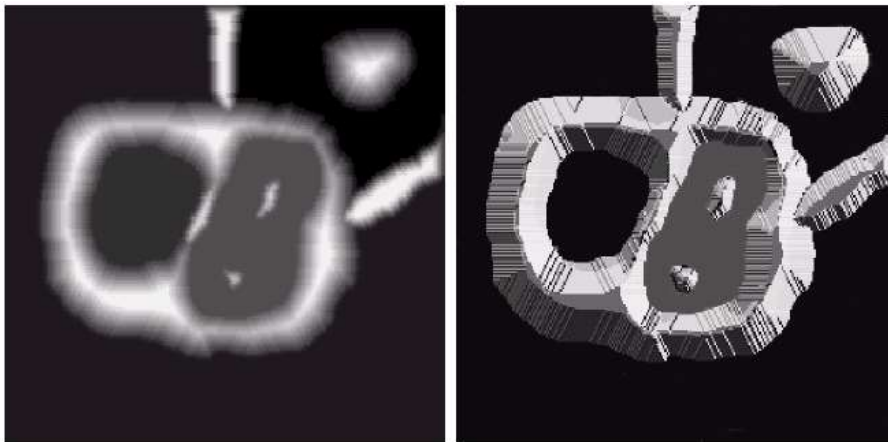
8-connectivity



Segmentation Using Morphological Watersheds

Three types of points in a topographic interpretation:

- Points belonging to a regional minimum
- Points at which a drop of water would fall to a single minimum. (→The *catchment basin* or *watershed* of that minimum.)
- Points at which a drop of water would be equally likely to fall to more than one minimum. (→The *divide lines* or *watershed lines*.)



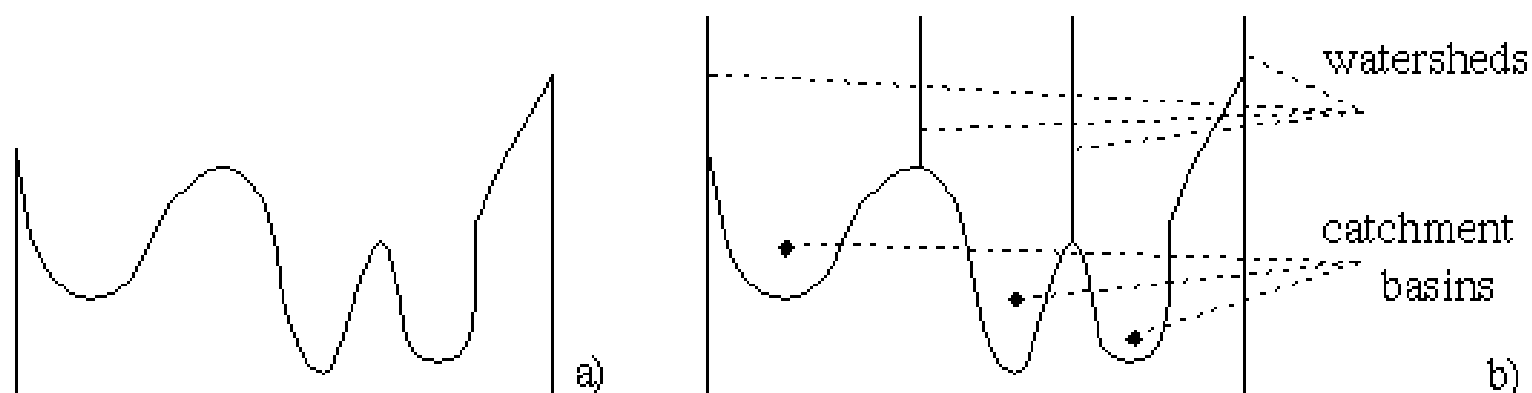
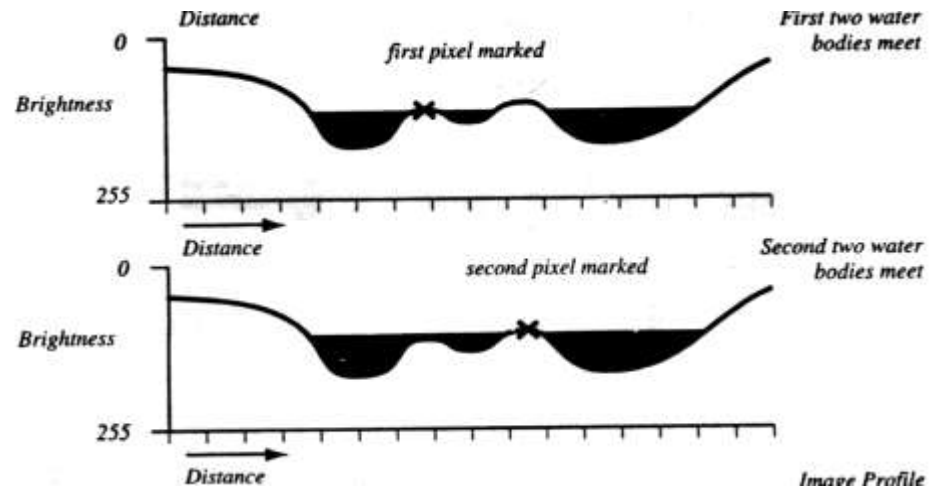
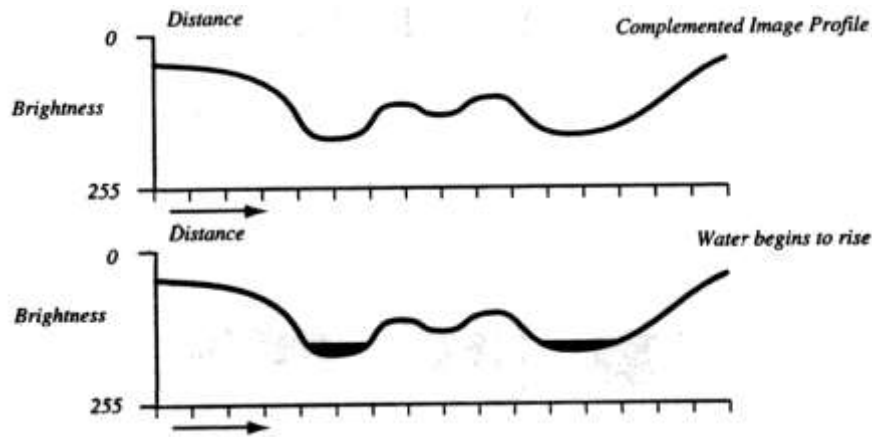


Figure 5.47 *One-dimensional example of watershed segmentation. (a) Gray level profile of image data. (b) Watershed segmentation – local minima of gray level (altitude) yield catchment basins, local maxima define the watershed lines.*



- ▶ The objective is to find watershed lines.
- ▶ The idea is simple:
 - Suppose that a hole is punched in each regional minimum and that the entire topography is flooded from below by letting water rise through the holes at a uniform rate.
 - When rising water in distinct catchment basins is about the merge, a dam is built to prevent merging. These dam boundaries correspond to the watershed lines.

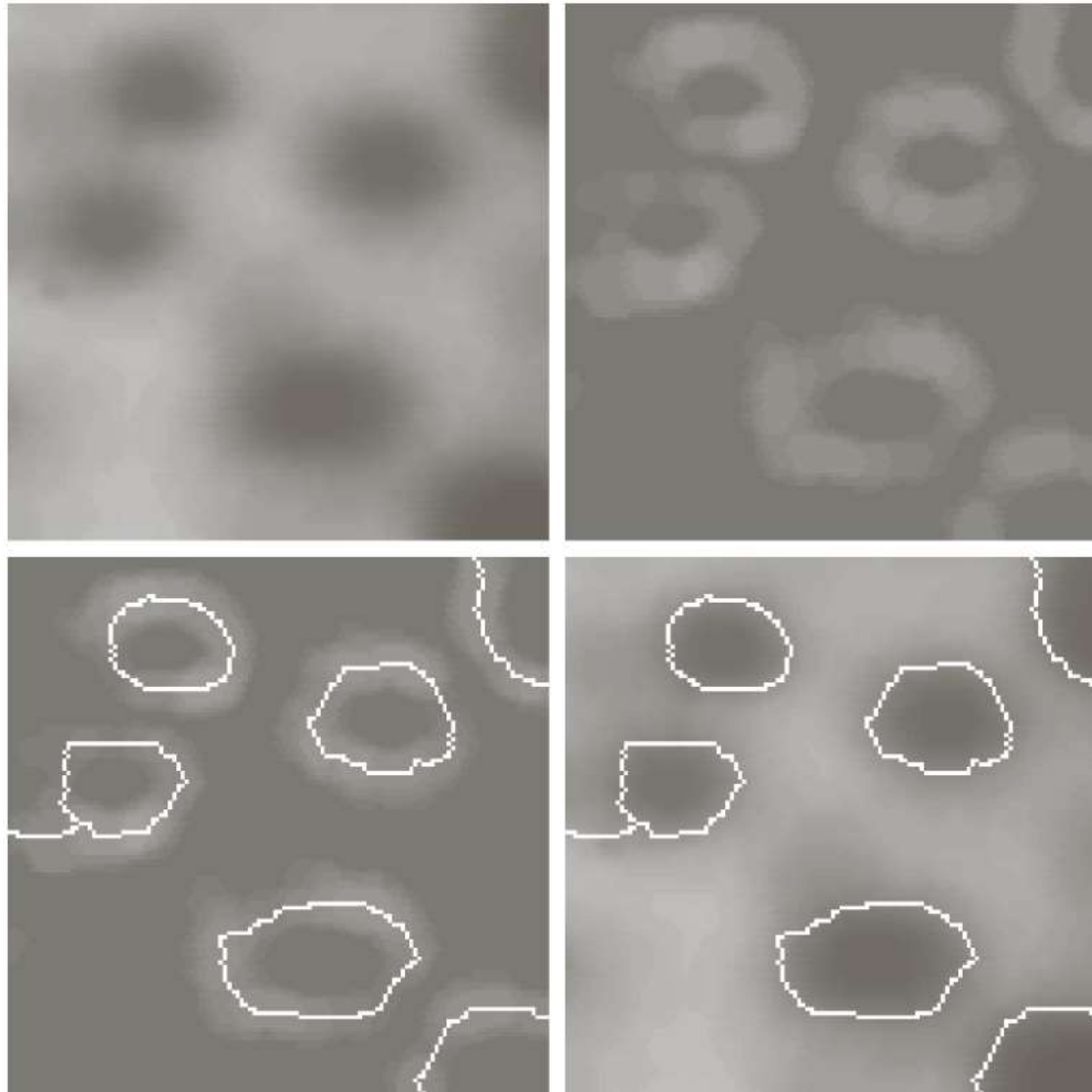




- ▶ Start with all pixels with the lowest possible value.
 - These form the basis for initial watersheds
- ▶ For each intensity level k :
 - For each group of pixels of intensity k
 1. If adjacent to exactly one existing region, add these pixels to that region
 2. Else if adjacent to more than one existing regions, mark as boundary
 3. Else start a new region



Watershed algorithm is often used on the gradient image instead of the original image.



a	b
c	d

FIGURE 10.56

(a) Image of blobs.
(b) Image gradient.
(c) Watershed lines.
(d) Watershed lines superimposed on original image.
(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)



K-means Clustering

- Partition the data points into K clusters randomly. Find the centroids of each cluster.
- For each data point:
 - Calculate the distance from the data point to each cluster.
 - Assign the data point to the closest cluster.
- Recompute the centroid of each cluster.
- Repeat steps 2 and 3 until there is no further change in the assignment of data points (or in the centroids).



Clustering

Example





Questions???

Thank You!!!